Prediction of validation dataset

The LOGISTIC Procedure

Model Information	
Data Set	WORK.LOGISTIC
Response Variable	dflt
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	132
Number of Observations Used	132

Response Profile		
Ordered Value	dflt	Total Frequency
1	1	14
2	0	118

Probability modeled is dflt=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	91.285	31.122
SC	94.167	51.302
-2 Log L	89.285	17.122

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	72.1625	6	<.0001
Score	76.7635	6	<.0001
Wald	9.3022	6	0.1573

Analysis of Maximum Likelihood Estimates

-2 Log L	89.285	17.122

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	72.1625	6	<.0001
Score	76.7635	6	<.0001
Wald	9.3022	6	0.1573

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-24.2386	8.0649	9.0327	0.0027
Utilisation	1	9.2090	4.8407	3.6191	0.0571
LTV	1	0.8409	0.5561	2.2870	0.1305
Borrowing_portfolio_	1	6.2435	4.7283	1.7436	0.1867
Postcode_index	1	0.0789	0.0505	2.4391	0.1183
arrears_flag	1	5.9412	2.5582	5.3937	0.0202
relationship_length	1	0.2609	0.2891	0.8145	0.3668

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Utilisation	>999.999	0.757	>999.999
LTV	2.319	0.780	6.895
Borrowing_portfolio_	514.641	0.049	>999.999
Postcode_index	1.082	0.980	1.195
arrears_flag	380.382	2.528	>999.999
relationship_length	1.298	0.737	2.287

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	98.8	Somers' D	0.977
Percent Discordant	1.2	Gamma	0.977
Percent Tied	0.0	Tau-a	0.187
Pairs	1652	c	0.988

Prediction of validation dataset

Prediction of validation dataset

The GENMOD Procedure

Model Information	
Data Set	WORK.OVERALL
Distribution	Binomial
Link Function	Probit
Dependent Variable	dflt
Scale Weight Variable	validation_sample

Number of Observations Read	143
Number of Observations Used	132
Sum of Weights	132
Number of Events	14
Number of Trials	132
Missing Values	11

Response Profile			
Ordered Value	dflt	Total Frequency	Total Weight
1	1	14	14
2	0	118	118

PROC GENMOD is modeling the probability that dflt='1'.

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Log Likelihood		-11.1309	
Full Log Likelihood		-11.1309	
AIC (smaller is better)		32.2619	
AICC (smaller is better)		32.7381	
BIC (smaller is better)		46.6759	

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	-9.9197	2.6782	-15.1689	-4.6705	13.72	0.0002

Algorithm converged.

Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	-9.9197	2.6782	-15.1689	-4.6705	13.72	0.0002
Utilisation	1	6.2753	2.2933	1.7806	10.7700	7.49	0.0062
Borrowing_portfolio_	1	3.8592	1.9509	0.0355	7.6829	3.91	0.0479
Postcode_index	1	0.0265	0.0221	-0.0168	0.0698	1.44	0.2302
arrears_flag	1	2.3629	0.7564	0.8805	3.8454	9.76	0.0018
Scale	0	1.0000	0.0000	1.0000	1.0000		

Note: The scale parameter was held fixed.

Prediction of validation dataset

Parameter	Estimate
Intercept	-9.919741832
Utilisation	6.2753186044
Borrowing_portfolio_	3.8591911182
Postcode_index	0.0265093379
arrears_flag	2.3629197893
Scale	1.0000000000

Prediction of validation dataset

The LOGISTIC Procedure

Model Information		
Data Set	WORK.PREDS	Predicted Values and Diagnostic Statistics
Response Variable	dflt	
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	11
Number of Observations Used	11

Messages: 6 User: u64366633

Number of Observations Used	11

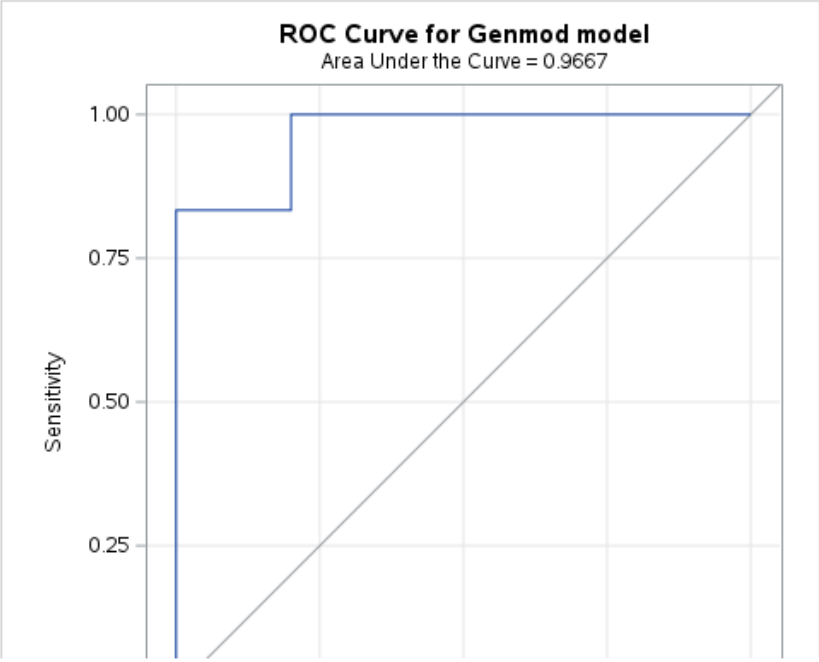
Response Profile		
Ordered Value	dflt	Total Frequency
1	1	6
2	0	5

Probability modeled is dflt=1.

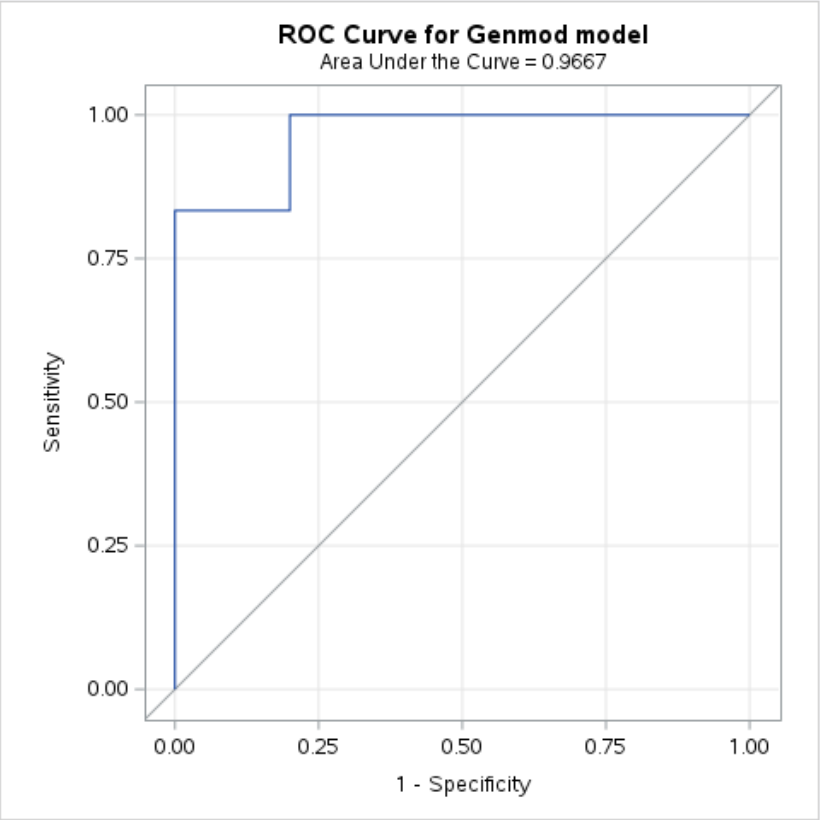
Score Test for Global Null Hypothesis		
Chi-Square	DF	Pr > ChiSq
0.0000	0	<.0001

ROC Model: Genmod model

ROC Model Information		
ROC Contrast Coefficients	pred	Predicted Value



ROC Contrast Coefficients	pred	Predicted Value



ROC Association Statistics							
	Mann-Whitney						
ROC Model	Area	Standard Error	95% Wald Confidence Limits		Somers' D	Gamma	Tau-a
Genmod model	0.9667	0.0471	0.8743	1.0000	0.9333	0.9333	0.5091

Prediction of validation dataset

The CORR Procedure

Prediction of validation dataset

The CORR Procedure

6 Variables:	Utilisation LTV Borrowing_portfolio_ratio Postcode_index arrears_flag relationship_length

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
Utilisation	143	0.61309	0.25509	87.67135	0.03000	1.89339	Utilisation
LTV	143	3.07404	6.88601	439.58826	0.07317	67.14433	LTV
Borrowing_portfolio_ratio	143	0.16772	0.16395	23.98463	0.00110	0.88403	Borrowing_portfolio_ratio
Postcode_index	143	74.89510	14.04054	10710	50.00000	99.00000	Postcode_index
arrears_flag	143	0.25175	0.43554	36.00000	0	1.00000	
relationship_length	143	5.96503	3.01735	853.00000	1.00000	9.00000	

Pearson Correlation Coefficients, N = 143
Prob > |r| under H0: Rho=0

	Utilisation	LTV	Borrowing_portfolio_ratio	Postcode_index	arrears_flag	relationship_length
Utilisation Utilisation	1.00000	0.28092 0.0007	0.43981 <.0001	0.02092 0.8041	0.20832 0.0125	0.11063 0.1884
LTV LTV	0.28092 0.0007	1.00000	0.22750 0.0063	-0.02823 0.7379	0.06764 0.4222	0.04657 0.5808
Borrowing_portfolio_ratio Borrowing_portfolio_ratio	0.43981 <.0001	0.22750 0.0063	1.00000	0.05569 0.5088	0.30987 0.0002	-0.02854 0.7351
Postcode_index Postcode_index	0.02092 0.8041	-0.02823 0.7379	0.05569 0.5088	1.00000	-0.00602 0.9432	0.07488 0.3741
arrears_flag	0.20832 0.0125	0.06764 0.4222	0.30987 0.0002	-0.00602 0.9432	1.00000	0.06033 0.4741
relationship_length	0.11063 0.1884	0.04657 0.5808	-0.02854 0.7351	0.07488 0.3741	0.06033 0.4741	1.00000